

Models.py

```
class HDD:
    def __init__(self, id, name, capacity_gb, computer_id):
        self.id = id
        self.name = name
        self.capacity_gb = capacity_gb
        self.computer_id = computer_id

class Computer:
    def __init__(self, id, name):
        self.id = id
        self.name = name

class CompHDD:
    def __init__(self, computer_id, hdd_id):
        self.computer_id = computer_id
        self.hdd_id = hdd_id
```

data.py

```
from models import HDD, Computer, CompHDD

computers = [
    Computer(1, 'Игровой ПК'),
    Computer(2, 'Офисный ПК'),
    Computer(3, 'Сервер'),
]

hdds = [
    HDD(1, 'Seagate Barracuda', 2000, 1),
    HDD(2, 'WD Red', 220000, 3),
    HDD(3, 'Kingston SSD', 500, 2),
    HDD(4, 'ADATA SSD Ultimate SU630', 240, 1),
    HDD(5, 'Samsung 970 Evo', 500, 3),
]

comps_hdds = [
    CompHDD(1, 1),
    CompHDD(1, 4),
    CompHDD(2, 3),
    CompHDD(3, 2),
    CompHDD(3, 5),
    CompHDD(3, 1),
    CompHDD(1, 3),
]
```

logic.py

```
from models import HDD, Computer, CompHDD

computers = [
    Computer(1, 'Игровой ПК'),
    Computer(2, 'Офисный ПК'),
    Computer(3, 'Сервер'),
]

hdds = [
    HDD(1, 'Seagate Barracuda', 2000, 1),
    HDD(2, 'WD Red', 220000, 3),
```

```

HDD(3, 'Kingston SSD', 500, 2),
HDD(4, 'ADATA SSD Ultimate SU630', 240, 1),
HDD(5, 'Samsung 970 Evo', 500, 3),
]

comps_hdds = [
    CompHDD(1, 1),
    CompHDD(1, 4),
    CompHDD(2, 3),
    CompHDD(3, 2),
    CompHDD(3, 5),
    CompHDD(3, 1),
    CompHDD(1, 3),
]

```

tests.py

```

import unittest
from data import computers, hdds, comps_hdds
from logic import (
    get_one_to_many,
    get_many_to_many,
    query_b1,
    query_b2,
    query_b3
)

class TestRK2(unittest.TestCase):

    def setUp(self):
        self.one_to_many = get_one_to_many(computers, hdds)
        self.many_to_many = get_many_to_many(computers, hdds, comps_hdds)

    def test_query_b1(self):
        """Проверка задания B1"""
        result = query_b1(self.one_to_many)
        self.assertEqual(
            result,
            [('ADATA SSD Ultimate SU630', 240, 'Игровой ПК')]
        )

    def test_query_b2(self):
        """Проверка задания B2"""
        result = query_b2(self.one_to_many, computers)
        self.assertEqual(
            result,
            [
                ('Игровой ПК', 240),
                ('Офисный ПК', 500),
                ('Сервер', 500)
            ]
        )

    def test_query_b3(self):
        """Проверка задания B3"""
        result = query_b3(self.many_to_many)
        self.assertEqual(result[0][0], 'ADATA SSD Ultimate SU630')
        self.assertEqual(result[-1][0], 'WD Red')

if __name__ == '__main__':
    unittest.main(verbosity=2)

```

main.py

```

from data import computers, hdds, comps_hdds
from logic import *

def main():
    one_to_many = get_one_to_many(computers, hdds)
    many_to_many = get_many_to_many(computers, hdds, comps_hdds)

    print("Задание B1")
    print(query_b1(one_to_many))

    print("\nЗадание B2")
    print(query_b2(one_to_many, computers))

    print("\nЗадание B3")
    print(query_b3(many_to_many))

if __name__ == '__main__':
    main()

```

Результаты тестирования:

```

test_query_b1 (__main__.TestRK2.test_query_b1)
Проверка задания B1 ... ok
test_query_b2 (__main__.TestRK2.test_query_b2)
Проверка задания B2 ... ok
test_query_b3 (__main__.TestRK2.test_query_b3)
Проверка задания B3 ... ok

```

```

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Ran 3 tests in 0.000s

```

```

OK

```

```

Process finished with exit code 0

```